HAIRUL AZHAR BIN ABU BAKAR

FORWARD-DEPLOYED AI ENGINEER | AI SOLUTIONS ARCHITECT | SYSTEMS FOUNDER

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THE ARCHITECTURAL NARRATIVE: Track record of an engineer bridging physical operational rigor with deep, sovereign AI architecture. With 10+ years scaling enterprise supply chains, luxury hospitality portfolios, and Fortune 500 logistics operations at 99.8% SLAs, I transitioned into elite AI engineering to solve unreliability, hallucinations, and thin wrapper syndrome. As a technical founder and forward-deployed architect, I build complete systems from the metal up—architecting custom knowledge-distilled model tiers (lucy, tara, stargazer, aim), deterministic safety governance (QGov), and multi-pass cross-model verification (K-Computer) that transform language models into trusted enterprise infrastructure.

LIVE PRODUCT ECOSYSTEM SHOWCASE

KatalockAi
katalock.net
Global agentic workspace featuring distilled LLM tiers, K-Computer verification, browser automation & live code sandbox.

Katalock Academy
katalock.net/academy
523-lesson AI engineering platform with Socratic tutor Professor Katalock reviewing student code in real time.

AIM OS
aimglobal.io
Sovereign agentic OS spanning 50+ integrated apps, 108 API endpoints, QGov safety engine & killswitch.

TECHNICAL SKILLS & ARCHITECTURE STACK

AI & Agentic Engineering:

Agentic AI | Knowledge Distillation | Model Quantization | Multi-Agent Orchestration | Cross-Model Verification | RAG | Model Context Protocol (MCP) | Anti-Hallucination Governance | Vector DBs (Pinecone, Chroma)

Full-Stack & Systems Infrastructure:

Python | TypeScript | React / Next.js | Node.js / Express | FastAPI / Django | PostgreSQL / Supabase | Redis | Docker / Kubernetes | AWS / GCP | Terraform

PROFESSIONAL EXPERIENCE: THE ARC OF EXECUTION

AI Platform Architect & Co-Founder | KatalockAi (katalock.net)

2026 – Present

Global Platform / Remote Workspace

- Own backend architecture, agent orchestration, and infrastructure end-to-end for a global agentic AI workspace while co-founder drives on-the-ground operations.
- Architected proprietary model engine utilizing **knowledge distillation processes**, establishing named tiers (katalock-lucy, katalock-tara, katalock-stargazer, katalock-aim) optimized for cost-performance and localized for India, Singapore, and Indonesia.
- Built advanced engineering surfaces, including supervised browser automation with safety pause gates, live workflow dependency graphs, evidence-ledger CRMs, and an in-workspace live code sandbox that builds, runs, publishes, and pushes to GitHub.

Architect & Founder | KatalockAi Academy (katalock.net/academy)

2026 – Present

Global Platform / Remote

- Designed and built a 20-phase, 523-lesson AI engineering platform featuring **Professor Katalock**, an embedded Socratic AI mentor reviewing student code in real time across six career routes.
- **Inference & Infrastructure Serving:** Authored deep-dive modules on production LLM serving (vLLM, SGLang, TensorRT-LLM), GPU autoscaling on Kubernetes (Karpenter, KAI scheduler), disaggregated prefill/decode chunking, and KV-cache optimization (RadixAttention, PagedAttention, LMCache offloading).
- **FinOps & Quantization Regimes:** Integrated low-bit quantization mechanics (AWQ, GPTQ, FP8, NVFP4), speculative decoding, model cascading, and unit-economic cost attribution models.
- **Safety, Alignment & Governance:** Structured comprehensive curriculum tracks on SFT, RLHF, DPO/IPO/KTO alignment families, reward hacking mitigations, Constitutional AI, and defenses against alignment faking.

PROFESSIONAL EXPERIENCE (CONTINUED)

Founder & AI Platform Architect | **Agentic Intelligence Matrix (AIM OS - aimglobal.io)** — Jan 2023 – Present
Singapore / Remote

- Architected AIM OS, a sovereign operating system for agentic AI spanning 50+ integrated apps, 108 API endpoints, 27+ autonomous agents, and ~80 database tables.
- Designed enterprise-grade governance, including the QGov policy engine, Sentinel safety monitor, and a founder killswitch for strict safety boundary adherence.
- Built multi-agent orchestration (Nexus Hub) with intent routing, pg-boss async job queue with automatic retries, and real-time offline-first collaboration using Yjs CRDTs.

Co-Founder & Operations Director | **Freda'D Pte Ltd / DGCC Investment Pte Ltd** — Apr 2014 – Dec 2024
Singapore / Middle East

- Scaled a bootstrapped fragrance venture from $500 to a profitable international enterprise with 25% YoY growth across luxury hotel portfolios in Asia and the Middle East.
- Deployed early automation and inventory systems directly into client operations, cutting overhead by 30% and boosting engagement by 40% at 99.8% SLAs.

Logistics Technology Operations | **GOGOVAN Singapore (GOGOX)** — Jul 2024 – Sep 2024
Singapore

- Managed enterprise logistics technology for IKEA Singapore, achieving 99.8% on-time delivery for 5,000+ daily orders and cutting incident resolution time by 25%.

Co-Founder (Investor Relations & GTM) | **Ignis Eco - Plastic-to-Fuel Pyrolysis (igniseco.com)** — 2026 – Present
Tamil Nadu / Bengaluru, India

- Co-founded a plastic-to-fuel pyrolysis venture, leading investor relations, seed-stage capital structuring, go-to-market strategy, and stakeholder outreach.

Project Director | **HH Design Pte Ltd / Interior Producer Pte Ltd** — Jan 2010 – Mar 2020
Singapore

- Delivered complex integrated-resort projects like Marina Bay Sands Casino and Resorts World Sentosa, managing budgets between $500K–$5M, improving timelines by 15%.

FEATURED PROJECTS & SYSTEMS

1. KatalockAi Workspace — katalock.net
Next.js, Supabase, Distillation, Docker, Stripe
Global agentic workspace featuring custom distilled LLMs (lucy, tara, stargazer, aim), K-Computer multi-pass verification, browser automation & code sandbox.

2. KatalockAi Academy — katalock.net/academy
Next.js, Supabase, Auto-Eval Engines, Dynamic Pedagogy
523-lesson AI engineering platform with Professor Katalock Socratic AI tutor providing real-time code reviews across serving, quantization & alignment.

3. AIM OS - Sovereign AI OS — aimglobal.io
Python, Node.js, Redis, K8s, Vector DBs
Sovereign AI operating system with governance layer (QGov), safety core (Sentinel), and multi-agent orchestration (Nexus Hub).

4. WorkflowGenius — github.com/CybergigSG/WorkflowGenius
Python, RAG, Pinecone, React, AWS Lambda
Enterprise RAG pipeline processing 10,000+ daily transactions at 99.5% uptime on AWS, cutting administrative workload by 40%.

EDUCATION & CREDENTIALS

- **MSc in Computer Science** – Heriot-Watt University *(Expected May 2028)*
- **Diploma in Architectural Technology** – Singapore Poly *(2001–2004)*

PROFESSIONAL CERTIFICATIONS

- **IBM Full Stack Developer Certificate** (2024) – React, Python, Docker, K8s
- **Google AI & Machine Learning Certificate** (2024) – GenAI, Data Science
- **Google Cybersecurity Professional Certificate** (2024)

ADDITIONAL INFORMATION

Languages: English (Native), Malay (Native)
Work Authorization: Singapore Citizen (Open to global relocation: USA, UK, EU, UAE, Australia, Canada, **India**, APAC).
Availability: Immediate / Two Weeks' Notice.